

Symmetric tensor powers reflect subnormality

Candidate solution packet for arXiv:1503.03692

13 August 2026

Status. Candidate full solution to the open question after Proposition 10.5 of Stochel–Stochel [1], pending expert review. The operator theorem below also answers the two-fold symmetric-tensor question left open by Stochel in 1996 [2].

1 Source question

The source is J. Stochel and J. B. Stochel, *Composition operators on Hilbert spaces of entire functions with analytic symbols*, arXiv:1503.03692v3 [1]. On PDF page 28, after Proposition 10.5, it asks whether the “only if” direction remains true when $\Phi'(0) = 0$. Here is the source passage, including the proposition to which the question refers.

If $\Phi'(0) \neq 0$, then we can also characterize subnormal composition operators.

PROPOSITION 10.5. *Suppose $\Phi \in \mathcal{F}$ is such that $\Phi'(0) \neq 0$. If $A \in \mathcal{B}(\mathcal{H})$, then C_A (resp., C_A^*) is subnormal if and only if A^* (resp., A) is subnormal.*

PROOF. This is a direct consequence of Theorems 7.2 7.4(i) and A.4(i), and the fact that if $T \in \mathcal{B}(\mathcal{H})$ is subnormal, then $T^{\odot n}$ is subnormal for every $n \in \mathbb{Z}_+$ (cf. [60] Theorem 2.4). $\square$

It is an open question as to whether the “only if” part of Proposition 10.5 remains true if $\Phi'(0) = 0$ (the “if” part is true independently of whether $\Phi'(0) = 0$ or not). If $\Phi(z) = z^2$ for $z \in \mathbb{C}$, then the question reduces to that in [60] p. 139].

Below we give necessary and sufficient conditions for positive operators $A, B \in \mathcal{B}(\mathcal{H})$ to satisfy the inequality $C_A \preccurlyeq C_B$.

In full: for a nonconstant entire function $\Phi(z) = \sum_{n \geq 0} a_n z^n$ with $a_n \geq 0$, and $A \in \mathcal{B}(\mathcal{H})$, does subnormality of C_A (respectively C_A^*) imply subnormality of A^* (respectively A) without assuming $a_1 = \Phi'(0) \neq 0$? For $\Phi(z) = z^2$, the source identifies this with the 1996 question: does subnormality of $T^{\odot 2}$ imply subnormality of T ?

2 Definitions and statement

Let P_m be the orthogonal projection from $\mathcal{H}^{\otimes m}$ onto its symmetric subspace $\mathcal{H}^{\odot m}$. For $T \in \mathcal{B}(\mathcal{H})$, the symmetric power $T^{\odot m}$ is the restriction of $T^{\otimes m}$ to $\mathcal{H}^{\odot m}$. A bounded operator is *subnormal* if it is the restriction of a normal operator to an invariant closed subspace.

Theorem 1 (Symmetric-power reflection). *Let T be a bounded operator on a complex Hilbert space and let $m \geq 1$. Then*

$$T^{\odot m} \text{ is subnormal} \iff T \text{ is subnormal.}$$

The forward implication is new relative to the two source questions. The reverse implication is standard: a normal extension of T induces a normal symmetric-power extension.

3 Proof intuition

Pure powers $x^{\otimes m}$ immediately descend hyponormality from $T^{\odot m}$ to T . If this already makes T normal, there is nothing more to prove. Otherwise a nonzero boundary point λ of the essential spectrum supplies a weakly-null approximate eigenvector sequence y_j . Insert an arbitrary vector x into one tensor slot and $m - 1$ copies of y_j into the others, then symmetrize. The k th orbit norm under $T^{\odot m}$ is a Stieltjes moment sequence in k . As $j \rightarrow \infty$, all cross terms vanish weakly and the remaining expression is

$$\frac{1}{m} |\lambda|^{2k(m-1)} \|T^k x\|^2.$$

Stieltjes moment sequences are closed under pointwise limits, and the harmless geometric factor can be rescaled away. Lambert's norm-moment criterion then forces T itself to be subnormal.

4 Proof

We use two standard facts. First, a bounded operator R is subnormal if and only if, for every x , the sequence $(\|R^k x\|^2)_{k \geq 0}$ is a Stieltjes moment sequence [2, Proposition 2.3]. Second, Stieltjes moment sequences are closed under pointwise convergence [2, (S2)].

Lemma 1 (A nonzero singular spectral point). *Let T be hyponormal on an infinite-dimensional Hilbert space. Either T is normal, or there exist $\lambda \neq 0$ and unit vectors $y_j \rightharpoonup 0$ such that*

$$\|(T - \lambda)y_j\| \rightarrow 0.$$

Proof. If the essential spectrum is $\{0\}$, then T is a Riesz operator, so its spectrum is countable. Putnam's inequality for hyponormal operators gives

$$\|T^*T - TT^*\| \leq \frac{1}{\pi} \text{Area}(\sigma(T)) = 0,$$

and T is normal.

Otherwise the compact set $\sigma_e(T)$ has a nonzero boundary point λ . A boundary point of the Fredholm essential spectrum cannot be an upper semi-Fredholm point: the upper semi-Fredholm set is open, and a non-Fredholm upper semi-Fredholm point has a whole neighborhood of non-Fredholm points (the extended index is locally constant). Hence $T - \lambda$ is not upper semi-Fredholm. The singular-sequence characterization of upper semi-Fredholm operators now gives unit vectors $y_j \rightharpoonup 0$ with $(T - \lambda)y_j \rightarrow 0$. $\square$

Lemma 2 (The symmetrized one-slot identity). *For $u, v \in \mathcal{H}$ and $m \geq 2$, put*

$$J_v u = P_m(u \otimes v^{\otimes(m-1)}) = \frac{1}{m} \sum_{r=1}^m v^{\otimes(r-1)} \otimes u \otimes v^{\otimes(m-r)}.$$

Then

$$\|J_v u\|^2 = \frac{1}{m} \|u\|^2 \|v\|^{2m-2} + \frac{m-1}{m} |\langle u, v \rangle|^2 \|v\|^{2m-4}. \quad (1)$$

Moreover,

$$(T^{\odot m})^k J_v u = J_{T^k v}(T^k u), \quad k \geq 0. \quad (2)$$

Proof. In the displayed sum defining $J_v u$, there are m diagonal inner products and $m(m-1)$ off-diagonal inner products. Their values are respectively $\|u\|^2\|v\|^{2m-2}$ and $|\langle u, v \rangle|^2\|v\|^{2m-4}$, which proves (1). Equation (2) follows because $T^{\otimes m}$ commutes with every tensor-slot permutation and hence with P_m . $\square$

Proof of Theorem 1. Only the forward implication needs proof. The case $m = 1$ is tautological. Put $S = T^{\odot m}$ and suppose $m \geq 2$ and S is subnormal. Then S is hyponormal. Applied to the pure symmetric tensor $x^{\otimes m}$, its hyponormality inequality gives

$$\|T^*x\|^m = \|S^*x^{\otimes m}\| \leq \|Sx^{\otimes m}\| = \|Tx\|^m.$$

Thus T is hyponormal. If $\mathcal{H}$ is finite-dimensional, tracing the self-commutator shows that every hyponormal operator is normal, and we are done. The same is true if T is normal in infinite dimension.

In the remaining case choose $\lambda \neq 0$ and y_j from Lemma 1. For each fixed k ,

$$T^k y_j - \lambda^k y_j \longrightarrow 0, \quad (3)$$

by the polynomial identity for $T^k - \lambda^k$. Fix $x \in \mathcal{H}$ and set $\xi_j = J_{y_j} x$. By subnormality of S and Lambert's criterion,

$$a_k^{(j)} := \|S^k \xi_j\|^2, \quad \text{quad } k \geq 0,$$

is a Stieltjes moment sequence. Lemma 2 and (2) give

$$\begin{aligned} a_k^{(j)} &= \frac{1}{m} \|T^k x\|^2 \|T^k y_j\|^{2m-2} \\ &\quad + \frac{m-1}{m} |\langle T^k x, T^k y_j \rangle|^2 \|T^k y_j\|^{2m-4}. \end{aligned}$$

By (3), $\|T^k y_j\| \rightarrow |\lambda|^k$. Also $T^k y_j \rightarrow 0$, so the second term tends to zero. Consequently, for every $k \geq 0$,

$$a_k^{(j)} \longrightarrow a_k := \frac{1}{m} |\lambda|^{2k(m-1)} \|T^k x\|^2. \quad (4)$$

The pointwise closure property makes (a_k) a Stieltjes moment sequence. Multiplication by m and removal of the positive geometric factor c^k , $c = |\lambda|^{2(m-1)}$, correspond respectively to scaling a representing measure and pushing it forward by $t \mapsto t/c$. Hence $(\|T^k x\|^2)_{k \geq 0}$ is a Stieltjes moment sequence. Since x was arbitrary, Lambert's criterion proves that T is subnormal. $\square$

5 Application to the source question

Corollary 1. *Let $\Phi(z) = \sum_{n \geq 0} a_n z^n$ be a nonconstant entire function with $a_n \geq 0$, and let $A \in \mathcal{B}(\mathcal{H})$. Without any hypothesis on $\Phi'(0)$,*

$$C_A \text{ is subnormal} \iff A^* \text{ is subnormal}, \quad C_A^* \text{ is subnormal} \iff A \text{ is subnormal}.$$

Proof. The Fock model of [1, Theorem 7.2] identifies C_A^* with the orthogonal sum of $\Xi_Q(A)^{\odot n}$ over n for which $a_n > 0$, where $\Xi_Q(A) = QAQ$ for a conjugation Q . Taking adjoints gives the corresponding model for C_A . Because Φ is nonconstant, at least one such index $n \geq 1$ exists. A reducing summand of a subnormal operator is subnormal, so Theorem 1, and invariance of subnormality under conjugation, give the two necessity directions. The sufficiency directions are the standard symmetric-power construction already recorded in [1, Proposition 10.5]; they never required $\Phi'(0) \neq 0$. $\square$

6 Verification

The companion script `code/verify_symmetrization.py` checked (1) and (2) in 400 seeded random complex cases with dimensions 2–4 and powers $m = 2$ –4. The worst relative error was below $2 \cdot 10^{-12}$. It also used the standard weakly-null approximate eigenvectors for a unilateral shift at a unimodular λ and observed the residual and the maximum error in (4) decrease over lengths 32, 64, $\dots$, 1024. The command was

```
conda run --no-capture-output -n sandbox python code/verify_symmetrization.py
```

and all checks passed. These finite computations check the tensor algebra and limiting formula; they do not prove the spectral lemma or subnormality.

7 Novelty check, limitations, and reviewer focus

The bounded check was performed on 13 August 2026. The run indexes were searched for arXiv:1503.03692, the exact title, “symmetric tensor power,” and “subnormality.” Web searches used the exact 1996 open-question wording, the 1996 DOI, the source question with $\Phi'(0) = 0$, and combinations of “subnormal” with “symmetric tensor square/power.” The recent survey Garcia–O’Loughlin–Yu, arXiv:2306.14807 [3], was also searched; it contains no occurrence of “subnormal.” Results located the original 1996 question and the 2017 source but no claimed solution. This is a bounded search, not a proof of novelty.

The theorem concerns bounded operators. It does not assert an unbounded symmetric-power reflection theorem. Corollary 1 allows the composition operator itself to be unbounded/subnormal in the source’s closed-operator sense; the extracted tensor summand is bounded because $A \in \mathcal{B}(\mathcal{H})$. A reviewer should check especially (i) the use of the upper-semi-Fredholm/singular-sequence characterization at a boundary point of $\sigma_e(T)$, (ii) the Riesz-hyponormal normality argument when $\sigma_e(T) = \{0\}$, and (iii) closure and rescaling of the limiting Stieltjes moment sequence.

References

- [1] J. Stochel and J. B. Stochel, *Composition operators on Hilbert spaces of entire functions with analytic symbols*, Journal of Mathematical Analysis and Applications **454** (2017), 1019–1066; arXiv:1503.03692.
- [2] J. Stochel, *Seminormality of operators from their tensor product*, Proceedings of the American Mathematical Society **124** (1996), 135–140. DOI: 10.1090/S0002-9939-96-03017-1.
- [3] S. R. Garcia, R. O’Loughlin, and J. Yu, *Symmetric and antisymmetric tensor products for the function-theoretic operator theorist*, Canadian Journal of Mathematics **77** (2025), 324–346; arXiv:2306.14807.